Define Problem Statements

Project: Rising Waters – Flood Prediction using Machine Learning
Team Lead: Atluri Jyothendra Sai Ram
Members:
• Battu Rama Rao
• Hima Varshini Bodapati
• Siddhardha Jilagam
• Udaya Teja

Purpose

This document explains the 'Define Problem Statements' stage of the Rising Waters project. The project predicts flood occurrence using historical weather data and machine learning. The documentation follows the SDLC flow from ideation to deployment.

Project Overview

Project Overview for this phase focuses on developing a reliable flood prediction solution. Historical rainfall, seasonal rainfall, and cloud visibility are analyzed to train classification models. This phase contributes to improving early warning, evacuation planning, disaster management, and decision making.

Objectives

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Dataset Description

Dataset Description for this phase focuses on developing a reliable flood prediction solution. Historical rainfall, seasonal rainfall, and cloud visibility are analyzed to train classification models. This phase contributes to improving early warning, evacuation planning, disaster management, and decision making.

Machine Learning Models Used (Decision Tree, Random Forest, KNN, XGBoost)

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Detailed Discussion

Flask Web Application

The Flask application integrates the best-performing model (XGBoost achieving about 96.55% accuracy). The application provides an easy interface for authorities to enter weather parameters and instantly receive flood risk predictions. Deployment on IBM Cloud improves scalability, accessibility, and future maintenance.

IBM Cloud Deployment Plan

The Flask application integrates the best-performing model (XGBoost achieving about 96.55% accuracy). The application provides an easy interface for authorities to enter weather parameters and instantly receive flood risk predictions. Deployment on IBM Cloud improves scalability, accessibility, and future maintenance.

Expected Outcomes

The Flask application integrates the best-performing model (XGBoost achieving about 96.55% accuracy). The application provides an easy interface for authorities to enter weather parameters and instantly receive flood risk predictions. Deployment on IBM Cloud improves scalability, accessibility, and future maintenance.

Team Contributions & Summary

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Responsibilities

Team Lead coordinated planning, integration, and reviews. Battu Rama Rao contributed to documentation, backend integration, and testing. Hima Varshini Bodapati assisted with requirement analysis and UI documentation. Siddhardha Jilagam supported model evaluation and testing. Udaya Teja worked on dataset preparation and deployment support.

Conclusion

The 'Define Problem Statements' document forms an important part of the complete project lifecycle. Together with the remaining documents it provides end-to-end evidence of planning, implementation, testing, documentation, and demonstration.